

SAJIN SAJI

Robotics & Automation Engineer | M.Eng Mechatronics & Cyber-Physical Systems

sajinsaji222@gmail.com | (+49) 017665883556 | Regensburg, Germany

linkedin.com/in/sajin-saji-4762561b5



PROFESSIONAL SUMMARY

Master's student in Mechatronics and Cyber-Physical Systems with a strong interest in systems engineering and automation. I have hands-on experience in system modeling, control systems, and integrating mechanical and electronic systems through academic projects and my role as a Junior Automation Engineer. I am particularly interested in applying my skills in aerospace and high-tech engineering, especially in system integration and testing.

TECHNICAL PROFILE

Programming & Tools: C++, Python, Unity, Git.

Simulation & Engineering Tools: MATLAB, Simulink, AutoCAD, SolidWorks.

Automation & Control Systems: Control Systems, Motion Control, Sensor Integration.

Cyber-Physical Systems: System Modeling, Digital Simulation.

Systems Engineering: System Integration, Verification & Validation (V&V), System Testing.

Engineering & Analysis: Data Analysis, System Performance Evaluation, Troubleshooting.

Operating Systems: Windows, Ubuntu.

Functional Safety: ISO 26262 (concepts and lifecycle).

Languages: English (B2), German (A2), Malayalam (Native).

AREAS OF INTEREST

Aerospace Systems, Avionics, System Integration, Automation, Simulation.

WORK EXPERIENCE

Junior Automation Engineer, Logix Space Technologies Pvt Ltd

07/2022 – 09/2023

- Developed and integrated automation systems combining mechanical, electronic, and control components, improving system performance and reliability.
- Implemented control solutions for automation processes to enhance system performance.
- Supported system testing, validation, and troubleshooting to ensure reliable operation.

Pandalam, India

Intern - Embedded System and Robotics, inFOX

07/2019 – 08/2019

- Worked on embedded systems programming and basic robotics applications.
- Applied hardware design concepts, including circuit prototyping and debugging.
- Performed testing and troubleshooting to ensure proper system functionality.

Kochi, India

EDUCATION AND TRAINING

MASTER OF ENGINEERING, MECHATRONIC AND CYBER-PHYSICAL SYSTEMS, Technische Hochschule Deggendorf (THD), Germany

10/2023 – Present

Relevant Courses: Advanced Robotics, Autonomous Systems, Cyber-Physical Systems, Additive Manufacturing, Human-Machine Interaction.

BACHELOR OF TECHNOLOGY, MECHANICAL ENGINEERING, APJ Abdul Kalam Technological University, Kerala, India

06/2018 – 06/2022

Relevant Courses: CAD and CAE, Material Science, Programming (C), Fluid and Thermal Engineering.

Grade: 2.4

PROJECTS

MASTER'S PROJECT, IMMERSIVE PEDESTRIAN SIMULATION WITH DYNAMIC VIRTUAL TRAFFIC (UNITY / AR/VR) 03/2025 – 06/2025

- Developed a Unity-based simulation environment modeling pedestrian behavior and dynamic traffic interactions.
- Implemented real-time movement logic, interaction handling, and basic collision behavior for realistic system dynamics.
- Simulated complex scenarios to analyse system behavior and improve interaction accuracy.
- Optimized simulation performance and usability through testing and iterative refinement.

MASTER'S PROJECT, INVOLUTE GEAR DESIGN WITH PROFILE SHIFT 03/2024 – 06/2024

- Analysed involute gear geometry and the effect of profile shift on tooth shape, meshing behaviour, and performance.
- Performed engineering calculations for tooth thickness, pitch, and center distance to evaluate gear characteristics.
- Developed a parametric gear model using IceSL and Lua scripting for design and analysis.

MASTER'S PROJECT, ADVANCEMENTS IN ELECTRET MICROPHONES 10/2023 – 01/2024

- Investigated electret condenser microphone (ECM) technology with focus on material behavior, miniaturization, and energy efficiency.
- Analysed polarization mechanisms, sensitivity, and performance characteristics to understand system behaviour.
- Evaluated ECM integration in IoT systems, audio devices, and medical instrumentation for real-world applications.

BACHELOR'S PROJECT, ANALYSIS OF THERMAL OPTIMIZATION OF ELECTRONIC COMPONENTS IN CIRCUITS 07/2021 – 06/2022

- Analysed thermal performance of electronic components in complex circuits.
- Simulated heat distribution and dissipation to evaluate system behaviour.
- Identified thermal hotspots and optimized heat sink design for improved efficiency.
- Contributed to improving circuit reliability through design modifications.

BACHELOR'S PROJECT, DESIGN AND ANALYSIS OF COMPRESSED AIR POWERED VEHICLE SYSTEM, *Mini Project* 08/2020 – 12/2020

- Investigated compressed air propulsion as a sustainable alternative to conventional fuel systems.
- Analysed system feasibility in terms of efficiency, cost, and environmental impact.
- Evaluated potential applications for low-emission and energy-efficient transportation.

LEADERSHIP EXPERIENCE

Student Chairman, Sree Buddha College of Engineering 2021 – 2022

Led the organization and execution of college-level technical events, including a tech fest.

CERTIFICATIONS & TRAINING

- Tele-Robotics - VHB (2025)
- Non-Destructive Testing (NDT) Certification (2021)
- AutoCAD 2D/3D Workshop (2021)
- Artificial Intelligence Internship, Goldmine (2021)

PERSONAL INTERESTS

Exploring new technologies, Simulation, Fitness and outdoor activities, Travelling.